Artificial intelligence

Evaluating the Future of Computing – Ethical and Security Challenges

The most material AI risks emerge during post-deployment operation, not design
Contestability as the organising principle

University of Essex

CASE STUD
GOVERNA
FAILURE

During operation, not during design or pre-launch

amazon

Recruitment

Welfare fraud

- **Amazon**: Trained on 10 years data → encoded gender bias → discontinued

- **SyRI:** Algorithmic fraud scoring → targeted lower-income areas → court invalidated

- **Critical insight:** Pre-deployment fairness checks were insufficient to predict systemic harm at scale
- **Root cause:** Historical patterns perpetuated through data selections
- **Systemic mechanism:** Feedback loops between historical data and automated optimisation

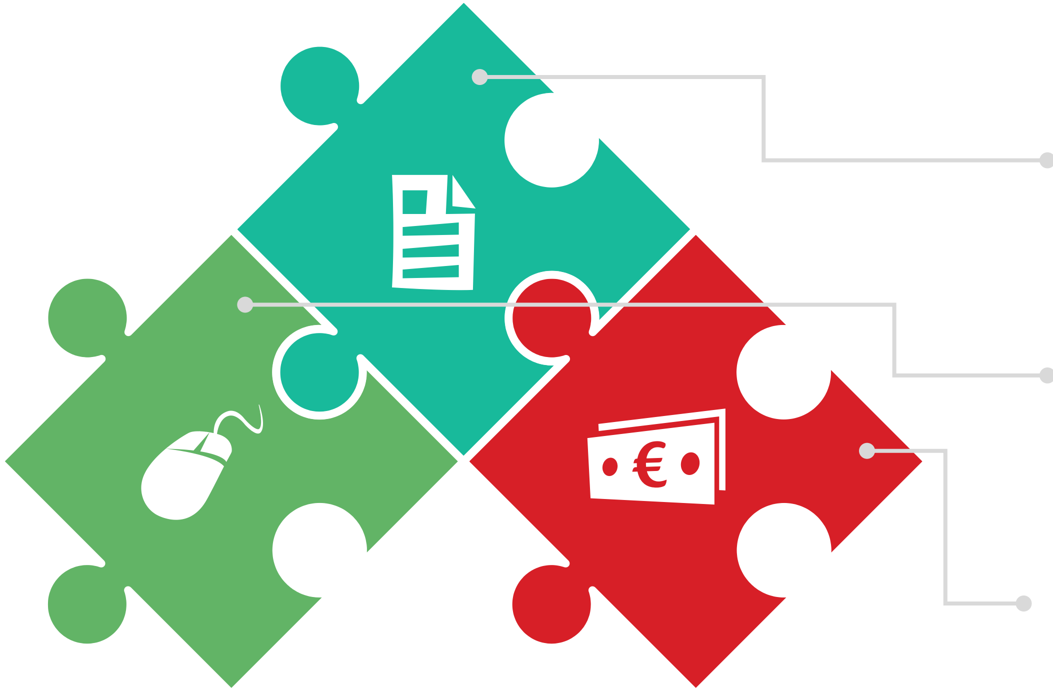
CITATIONS

✓ (Dastin, 2018) ✓ (District Court of The Hague, 2020)

REGULATORY LANDSCAPE

KEY: Complementary, not duplicative

Often embedded within existing enterprise cyber governance structures



EU AI ACT

BINDING
Extends GDPR
✓ ENFORCEMENT

NIST RMF

VOLUNTARY
4 Functions

OECD

PRINCIPLES
Lifecycle
Accountability

Limitation: Most frameworks insufficiently address behavioural change after deployment

CITATIONS

✓ (European Union, 2024) ✓ (NIST, 2023) ✓ (OECD, 2024) ✓ (Sousa e Silva, 2024)

SECURITY VULNERABILITIES AND TECHNICAL MITIGATION

These COMPLEMENT governance & require continuous monitoring.

THREATS

Adversarial inputs
mislead classifiers

Data poisoning
corrupt training data

Model extraction
steal parameters

Human-centred manipulation
Arup (2024):
Deepfake → $25M lost

COUNTER MEASURES

Differential privacy

Adversarial training

Federated learning

Human-in-the-loop controls

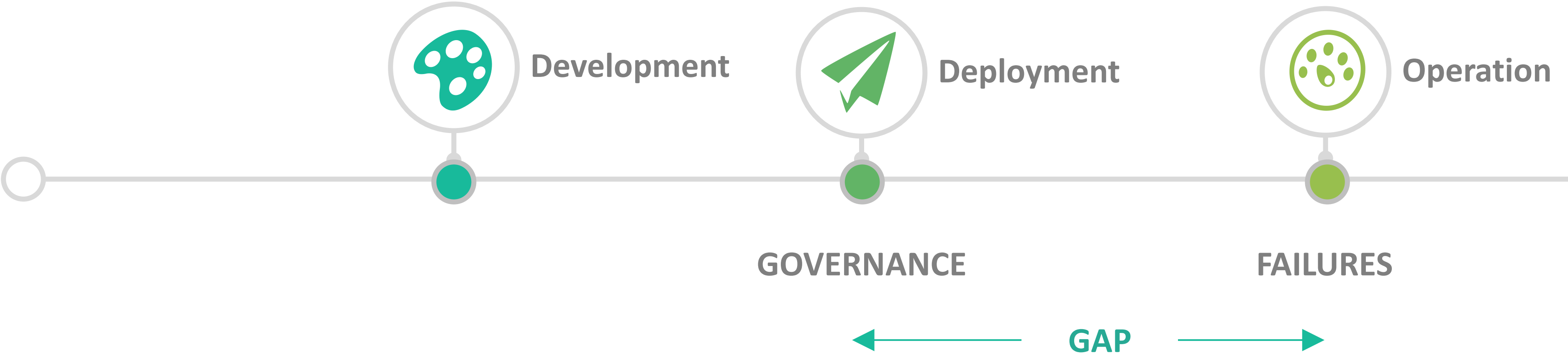
Identity authentication safeguards

Technical controls mitigate threats; they do not eliminate governance risk.

CITATIONS

✓ (Chan, 2024) ✓ (Deloitte Center for Financial Services, 2024) ✓ (Fu, Hadid and Damer, 2025) ✓ (Li and Zhang, 2025) ✓ (Salem et al., 2024)

THE GOVERNANCE GAP



- **Traditional:** hazards catalogued BEFORE deployment
- **Adaptive AI:** behaviours emerge DURING operation
- **Gap:** Oversight at launch ≠ Failures during operation
- **Proposition:** Governance and monitoring must be continuous, adaptive, and operationally embedded.
- **Enterprise impact:** Legal exposure, remediation cost, delayed adoption
- **Distribution shift:** training ≠ operation

CITATIONS

✓ Tamang (2025)

HEALTHCARE PARADOX AND ENTERPRISE STRATEGIES

These are competitive differentiators, not compliance burdens

HEALTHCARE PARADOX

More Governance = Higher Adoption

Counterintuitive

Demanding governance → Higher adoption success

Why?

Regulatory confidence + Clinical evidence + Public acceptance

ENTERPRISE PRACTICE

Microsoft: RAI Standard
Google: Model Cards
= Competitive Edge

Microsoft

Responsible AI Standard throughout product development

Google

Model Cards documenting behaviour across subgroups

Interpretation

Governance reduces uncertainty, accelerating enterprise adoption and market access

CITATIONS

✓ Nouis et al. (2025)

DEMONSTRATION – GOVERNANCE IN PRACTICE

Purpose: Translating governance principles into operational controls.
RAG system with built-in governance

RAG FEATURE

Drift threshold
2% - Detected via live output distribution monitoring

Audit logs
85% confidence minimum

Human Review
4-hour

FRAMEWORK PILLAR

Adaptive Structures
Triggers alerts

Accountability
Halts processing

Contestability
In practice

CITATIONS

✓ European Union (2024)

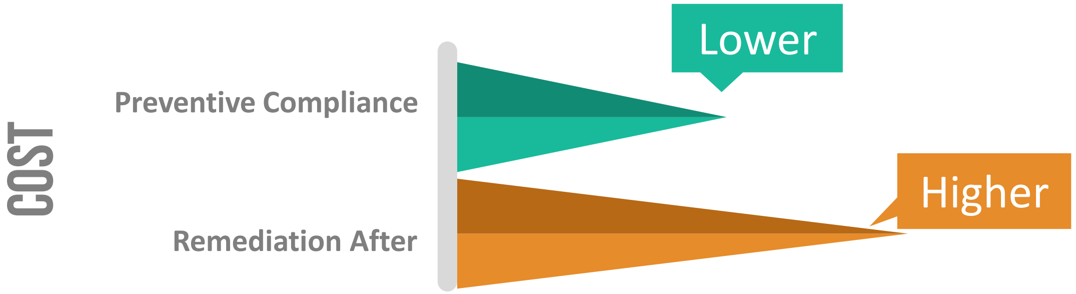
COUNTER-ARGUMENT AND REBUTTAL

- Governance hampers innovation
- Costs exceed risks

- Governance as organisational learning
- Monitoring → competitive intelligence

EVIDENCE: UNGOVERNED = MORE EXPENSIVE

Amazon: wasted investment + reputational harm
SyRI: judicial invalidation
Healthcare under oversight: successful commercialisation



EVALUATION

In many enterprise cases, governance costs are lower than systemic failure costs

CITATIONS

✓ (Saeri et al., 2025)

INTEGRATED GOVERNANCE FRAMEWORK (SYNTHETIC MODEL)

Adaptive structures

Continuous monitoring, not snapshots

Layered security

Technical + procedural controls

Accountability

Dataset scrutiny, metrics, explanations

ORGANISING PRINCIPLE

Contestability

CITATIONS

✓ Scaramuzza et al. (2025)

CONCLUSION

The challenge is not regulatory absence, but operational calibration.

- Transition toward responsive, operationally embedded governance
- Contestability as organising principle
- Equilibrium:
  Oversight cost < Anticipated losses
- Amazon/SyRI:
  Remediation > Compliance investment
- The frameworks exist.
  The question is calibration.

REFERENCES: Chan (2024); Dastin (2018); Deloitte Center for Financial Services (2024); District Court of The Hague (2020); European Union (2024); Fu et al. (2025); Li and Zhang (2025); Scaramuzza et al. (2025); NIST (2023); Nouis et al. (2025); OECD (2024); Saeri et al. (2025); Salem et al. (2024); Sousa e Silva (2024); Tamang (2025)